Fu Zeyuan | Mobile No.: +65 8399 6119 | Email: fuzeyuan2000@gmail.com | GitHub: github.com/Bardbar1an

EEE undergraduate with hands-on machine learning, LLM fine-tuning, and real-time systems experience, focused on building efficient, on-device AI solutions.

EDUCATION

Nanyang Technological University, Singapore Aug 2023 - Jun 2027
Bachelor of Engineering (Electrical and Electronics Engineering)

- Relevant Modules: (1) Circuit Analysis, (2) Digital Electronics, (3) Semiconductor Fundamentals, (4) Introduction to Data Science and AI, (5) Data Structures and Algorithms

PROJECTS

Personal Project: Local Large Language Model May 2024 - Present

- Fine-tuned a locally deployed large language model (LLM) to serve as an AI Dungeon Master for Dungeons & Dragons sessions, running fully on-device with no cloud API dependency and improving narrative flexibility and player immersion
- Curated and preprocessed D&D-specific datasets (rulebooks, campaign modules, and lore), training the model for genre-appropriate storytelling and consistent rule adherence.
- Iterated on fine-tuning data and prompts based on real player feedback to improve storytelling quality

FYP: Intelligent Control Design and Evaluation for a Mobile Ground Robot to Follow a Human Aug 2026 - Present

- Modeled a ground robot and its working environment for human-following scenarios.
- Designed, analyzed, and simulated an intelligent tracking control method combining Fuzzy Logic and Neural Networks for the robot to follow a human, implemented in the Webots simulation environment.
- Evaluated the developed control algorithms through simulation, analyzing tracking performance and robustness.

NTU-EEE Module: Introduction to Data Science and AI Aug 2024 - Oct 2024
Team Leader of 3

- Built a Python machine learning pipeline with scikit-learn to clean, train, and evaluate models on a Kaggle cardiovascular disease dataset
- Evaluated multiple models and identified the Decision Tree as the best performer, achieving 80% prediction accuracy
- Was able to resolve conflicts between group members to allow better teamwork and communications thus ensuring the deadline was met

Multi-Agent Path Finding (MAPF) Jan 2026 – May 2026

- Built an interactive Multi-Agent Path Finding (MAPF) simulation playground in Python/Pygame, featuring a resizable real-time render, world-to-screen coordinate mapping, and an in-app editor toggle for designing maps when the simulation is paused.
- Implemented a modular control loop that integrates a centralized planner/controller with a simulation engine, supporting a variable number of robots, start/reset/speed controls, time-stepped updates (dt), and live path visualization by rendering controller-computed paths each frame.

NTU-EEE Module: Data Structures and Algorithms Aug 2024 - Oct 2024

- Completed rigorous training in Python OOP + core data structures (stacks/queues, trees/heaps, BSTs) and algorithm analysis (growth rates, basic recurrences), applying concepts through weekly tutorials and graded assessments.
- Developed strong foundations in classic algorithms: sorting (merge sort, heapsort), searching (binary search), graph traversal (BFS/DFS), and greedy graph algorithms (Kruskal's MST, Dijkstra's shortest path), with exposure to basic AI classifiers/text classification.

EXPERIENCE

National Service Feb 2021 - Jan 2023
Electrical Technician

- Installed, maintained, and repaired Signal systems, ensuring 100% operational readiness for mission-critical operations.
- Adapted to high-pressure scenarios (e.g. emergency repairs during field exercises), demonstrating composure and problem-solving skills in resource-limited settings.

Shinryo Corporations Singapore Jul 2025 - Dec 2025
Electrical Intern

- Prepared and QA'd tender data: equipment schedule data entry + validation/standardization in Excel, using formulas like XLOOKUP/COUNTIF, plus templates/macros to automate repetitive checks and reduce manual errors.

- Supported cost/tender submissions: did cable-containment quantity take-offs from drawings and helped draft/update Interconnection Definition Documents (IDDs)—defining system interfaces, control points, and alarm states, coordinating with engineers/BIM to align with Integrated Digital Delivery (IDD) standards.

RELEVANT SKILLS

- Languages: English (fluent), Chinese (Mandarin) (fluent)
- Programming: Python, C++, Linux
- ML/AI: data preprocessing, model evaluation, scikit-learn, LLM fine-tuning
- Tools: Git/GitHub, Pygame, Webots, Microsoft Office (Excel, Word, PowerPoint)
- Hardware: circuit analysis, soldering, systems maintenance and configuration